Project Status Update and Forward Roadmap

Project: Quasar-Based Diagnostics of Gaia DR3 Parallax Systematics
Date: 2025-12-24
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Status: Active — Major Revision Phase

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1. Current Status Summary

This project constructs HEALPix maps of median Gaia DR3 quasar parallaxes to diagnose spatially varying parallax systematics. The methodological stance is explicitly conservative: the maps are framed as *diagnostic uncertainty fields* rather than predictive correction models.

Core components currently implemented and validated include:

- Median-based HEALPix aggregation at Nside=32 ($\sim 1.8^\circ$)
- Bootstrap uncertainty estimation per pixel
- Low-order ecliptic-latitude proxy model
- Spatially disjoint holdout validation using coarse HEALPix regions
- Quantification of spatial signal attenuation after conditioning on depth covariates
- Clear separation of descriptive (training-sky) structure from predictive generalization

The manuscript has undergone a structural and clarity revision, reducing length and sharpening the diagnostic framing. However, several *essential quantitative elements* were removed or left incomplete during revision and must be restored for scientific completeness and evaluability.

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2. Identified Deficiencies (Blocking Issues)

The following issues currently prevent external evaluation or submission:

2.1 Missing Quantitative Validation

- Test-set performance metrics (wRMSE) under spatial holdout are not reported.
- The empirical basis for the “diagnostic vs. predictive” distinction is therefore not verifiable.

2.2 Missing Coverage Metadata

- Sky coverage fraction after masking is not reported.
- Distribution of per-pixel source counts is not reported.

- Users cannot assess applicability or heteroscedasticity of the maps.

2.3 Proxy Model Performance Not Quantified

- No numerical measure of variance explained by the ecliptic-latitude proxy is provided.
- Claims that the proxy captures a “substantial fraction” of global structure are qualitative only.

2.4 Figure Redundancy

- Residual and raw median maps are visually similar at the chosen resolution.
- One figure duplicates another at vector resolution without adding information.

2.5 Selection and Contamination Sensitivity

- Dependence on the Gaia DR3 quasar candidate catalog is acknowledged but not stress-tested.
- No sensitivity check using a cleaner external catalog is provided.

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3. Roadmap Overview

The project is approximately **70% complete**. The remaining **30% consists primarily of reporting, validation, and contextualization**, not new methodological development.

The roadmap is divided into six sequential phases. Each phase has a clear exit criterion.

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4. Phase-by-Phase Plan

Phase 1 — Restore Empirical Falsifiability (Highest Priority)

Goal: Make the diagnostic framing empirically testable.

Actions:

- Restore a validation table reporting test-set wRMSE under spatial holdout for:
    - Zero baseline
    - Proxy model only
    - Proxy + residual field
- Report mean ± standard deviation across spatial splits.
- Explicitly state whether residuals fail to improve predictive performance.

Exit Criterion:
A skeptical reader can independently verify that residual maps do not generalize predictively.

Phase 2 — Restore Coverage and Sampling Metadata

Goal: Provide essential usage metadata.

Actions:

- Report sky coverage fraction after masking.
- Report median and 10–90% range of per-pixel source counts.
- Report fraction of sky below Nmin.

Exit Criterion:
Downstream users can assess applicability and limitations of the diagnostic maps.

Phase 3 — Quantify Proxy Model Performance

Goal: Replace qualitative language with numerical bounds.

Actions:

- Report variance explained (or equivalent metric) by the ecliptic-latitude proxy on the training sky.
- Report mean absolute parallax residual before and after proxy subtraction.
- Retain explicit statement that proxy is not physically complete.

Exit Criterion:
“Substantial fraction” is replaced by a bounded numerical statement.

Phase 4 — Resolve Figure Redundancy

Goal: Ensure each figure conveys unique information.

Actions (choose one):

- Replace the residual map with a difference map (median minus proxy).
- Or remove the residual figure entirely if differences are negligible at this scale.
- Remove duplicate vector-only figures unless required by journal format.

Exit Criterion:
No two figures communicate the same content.

Phase 5 — Selection and Contamination Sensitivity Check

Goal: Address the strongest external skepticism with minimal scope expansion.

Actions:

- Recompute a median parallax map using a cleaner external quasar catalog (e.g., CatNorth) or a stricter subset.
- Compare morphology and amplitude qualitatively or via RMS difference.
- Report whether major structures persist or attenuate.

Exit Criterion:
Evidence exists that major features are not driven solely by catalog contamination.

Phase 6 — Minimal External Contextualization

Goal: Situate the work within existing Gaia literature without competition claims.

Actions:

- Add a short comparison paragraph referencing:
    - Gaia DR3 global parallax zero-point
    - One independent estimate (e.g., quasar- or binary-based)
- Explicitly state that the work does not replace official corrections.

Exit Criterion:
Readers can contextualize amplitudes and limitations relative to known results.

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5. External Feedback Strategy (Post-Phase 4)

After Phases 1–4 are complete, targeted feedback may be solicited from 2–3 mid-career astronomers with experience in Gaia systematics.

The request should:

- Ask one specific methodological question
- Emphasize diagnostic framing
- Avoid requests for endorsement or collaboration

This step is intended for refinement, not validation rescue.

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6. Intended End State

Upon completion of Phases 1–6, the project will constitute:

- A falsifiable, conservative diagnostic contribution
- Suitable for submission as a Research Note or short Letter
- Explicitly bounded in scope and claims
- Reproducible and interpretable by downstream users

The work will not claim predictive correction capability but will provide empirically grounded uncertainty diagnostics for Gaia DR3 analyses.
